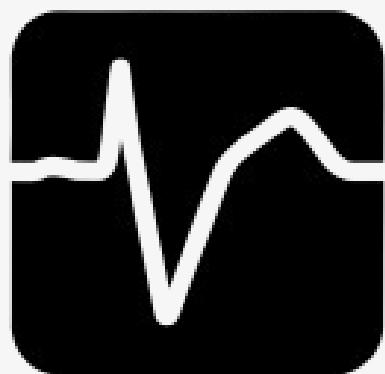
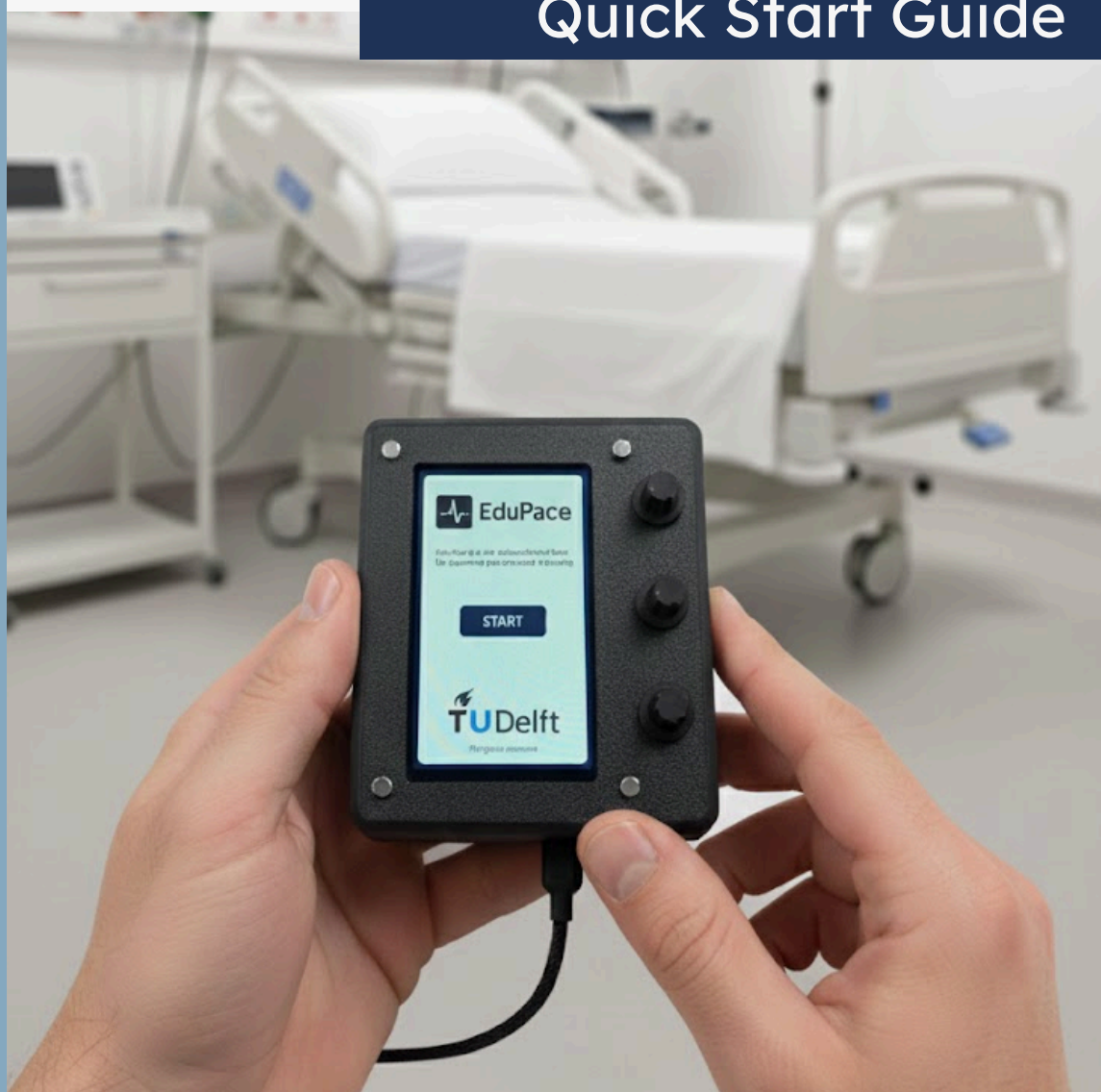




EduPace

User Manual

Quick Start Guide





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What is EduPace?

EduPace is an educational training tool designed for practicing the use of an external pacemaker. It has been developed by university students from TU Delft for educational purposes, specifically to support the training of nurses and clinicians.

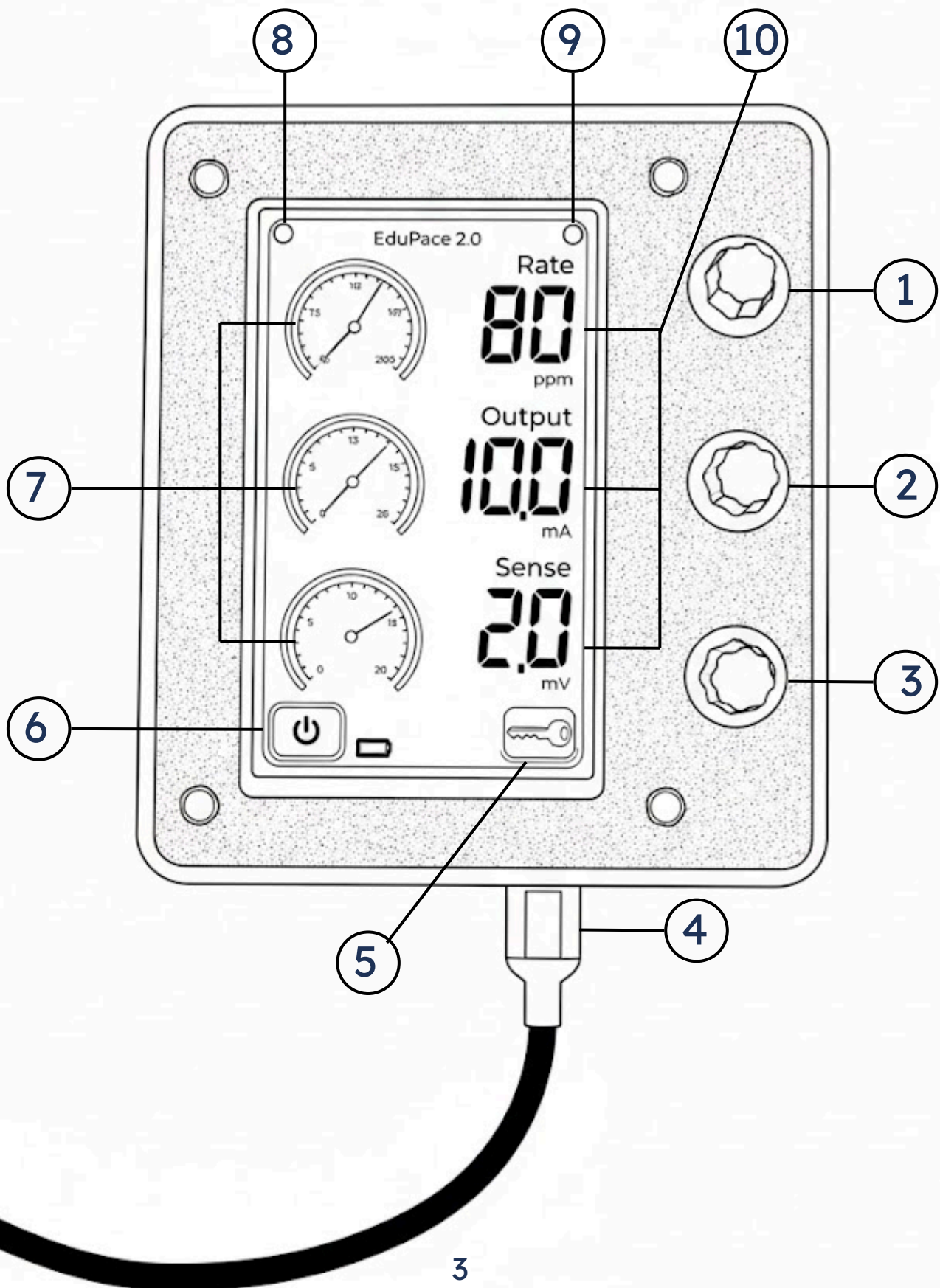
Reference Device

The design and functionality of EduPace are based on the Medtronic 53401 External Pacemaker, which was used as a reference during development.

Disclaimer

EduPace is not a medical device and is not intended for clinical use. It is an educational exercise tool only and must not be used for diagnosis, treatment, or patient care. Always use certified medical devices in clinical settings.

Controls and Indicators



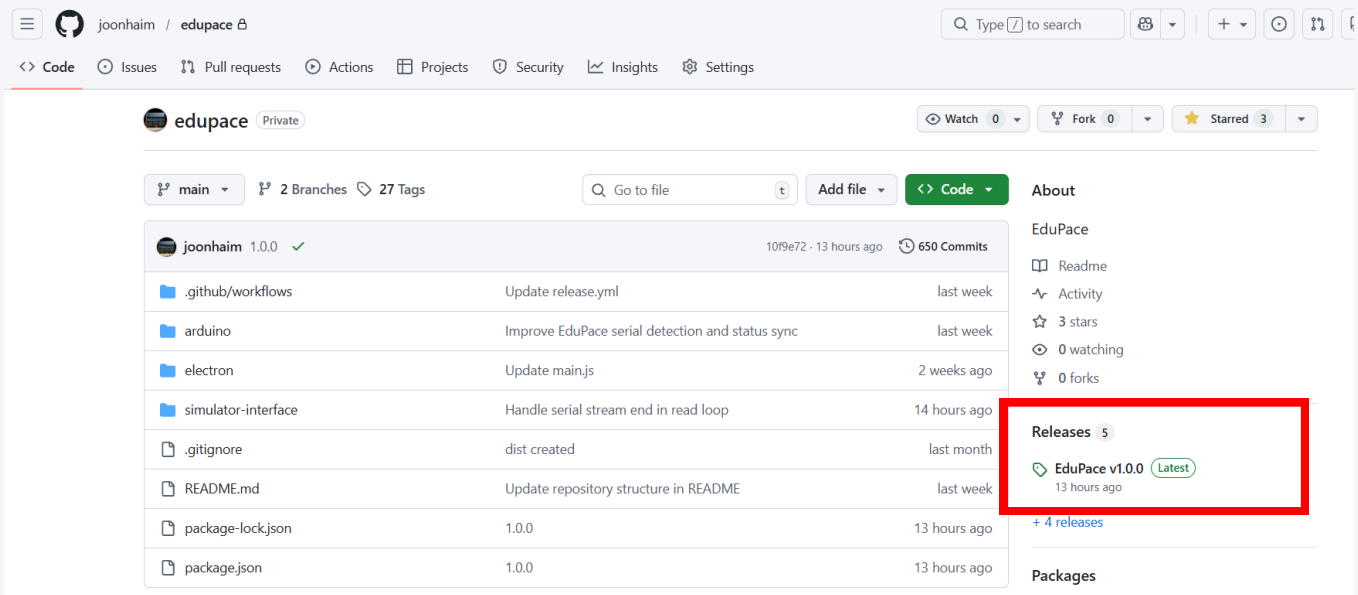
Controls and Indicators

1. RATE dial
2. OUTPUT dial
3. SENSE dial
4. USB-C input
5. Touchable Lock/Unlock button
6. Touchable On/Off button
7. Dashboard for RATE, OUTPUT, SENSE
8. Pacing status indicator LED
9. Sensing status indicator LED
10. Numeric indicators for RATE, OUTPUT, SENSE

Software Installation

1. Open a web browser and go to
<https://github.com/joonhaim/edupace.git>

2. Click on Releases on the right side of the page as shown below.



The screenshot shows the GitHub repository page for `joonhaim/edupace`. The repository is private and has 2 branches and 27 tags. The main branch is selected. The file list on the left includes `.github/workflows`, `arduino`, `electron`, `simulator-interface`, `.gitignore`, `README.md`, `package-lock.json`, and `package.json`. The right sidebar shows the 'Releases' tab selected, displaying 'EduPace v1.0.0' as the latest release, uploaded 13 hours ago. The 'About' section on the right lists the repository's README, activity, 3 stars, 0 watchers, and 0 forks. The 'Packages' section is also visible at the bottom.

System Installation

3. Download the appropriate installation file for your device:

Windows: Click **EduPace-1.0.0-win-x64.exe**

macOS: Click **EduPace-1.0.0-mac-arm64.dmg**

13 hours ago
github-actions
v1.0.0
10f9e72
Compare

EduPace v1.0.0 Latest

Release date: 21 January 2026
Platform: Desktop (Windows & macOS)

EduPace v1.0.0 is the first official desktop release of EduPace, a training simulator for temporary external ventricular pacing. The application is designed to work in combination with the dedicated **EduPace device**.

macOS Note

On macOS, it may be necessary to run the following Terminal command once to allow the app to launch due to Gatekeeper security restrictions:

```
sudo xattr -cr /Applications/EduPace.app
```

You will be prompted to enter your administrator password.
After running this command, the application should open normally.

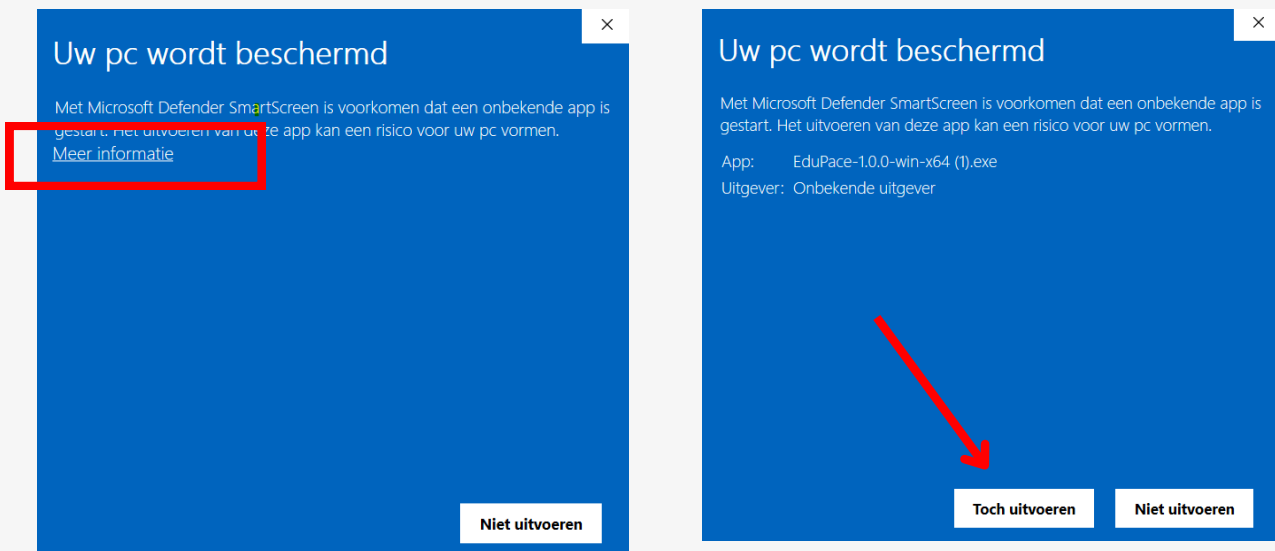
Assets 4

EduPace-1.0.0-mac-arm64.dmg	sha256:0ff768b80aaa88b3f...		101 MB	13 hours ago
EduPace-1.0.0-win-x64.exe	sha256:60a91a941a1f1dfcc...		83.2 MB	13 hours ago
Source code (zip)				13 hours ago
Source code (tar.gz)				13 hours ago

System Installation

4. Open the downloaded EduPace file to begin installation.

Windows users: If a security warning appears, click More info, then select Run anyway (“Toch uitvoeren”). This is safe.



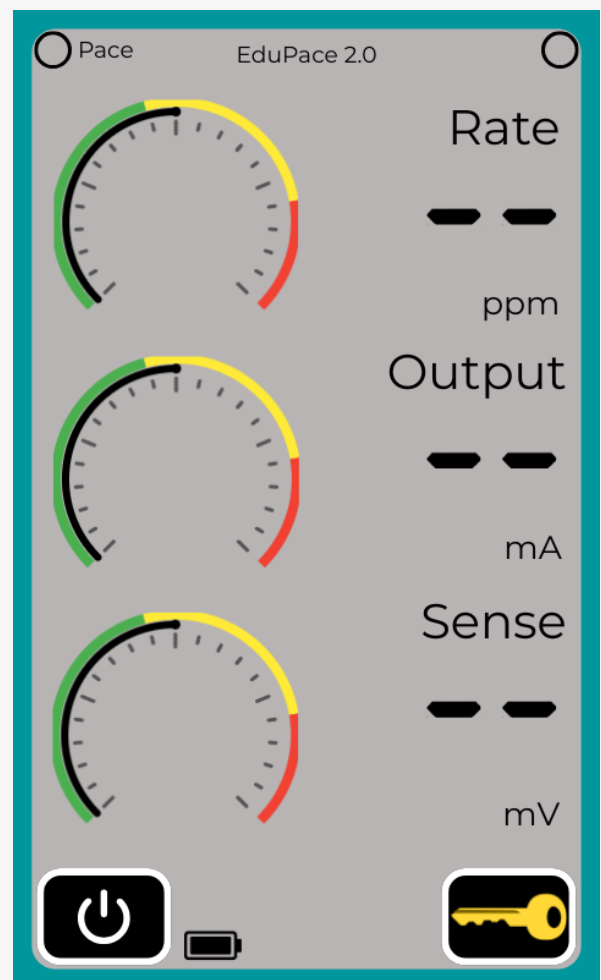
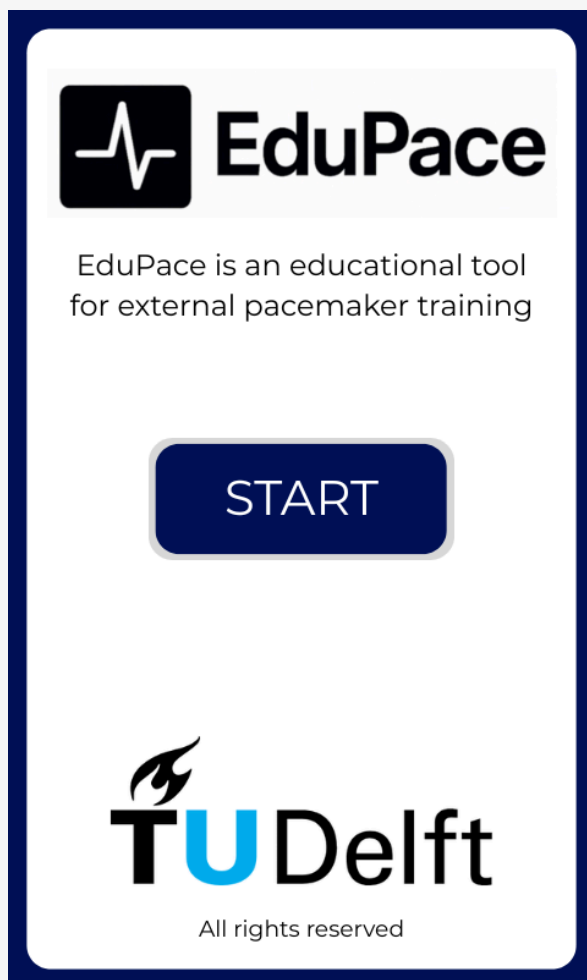
5. The installation wizard will appear. Click Next, then select Install to complete the installation process.

6. EduPace is now installed on your computer. You can now start using the software. Look for the following icon:



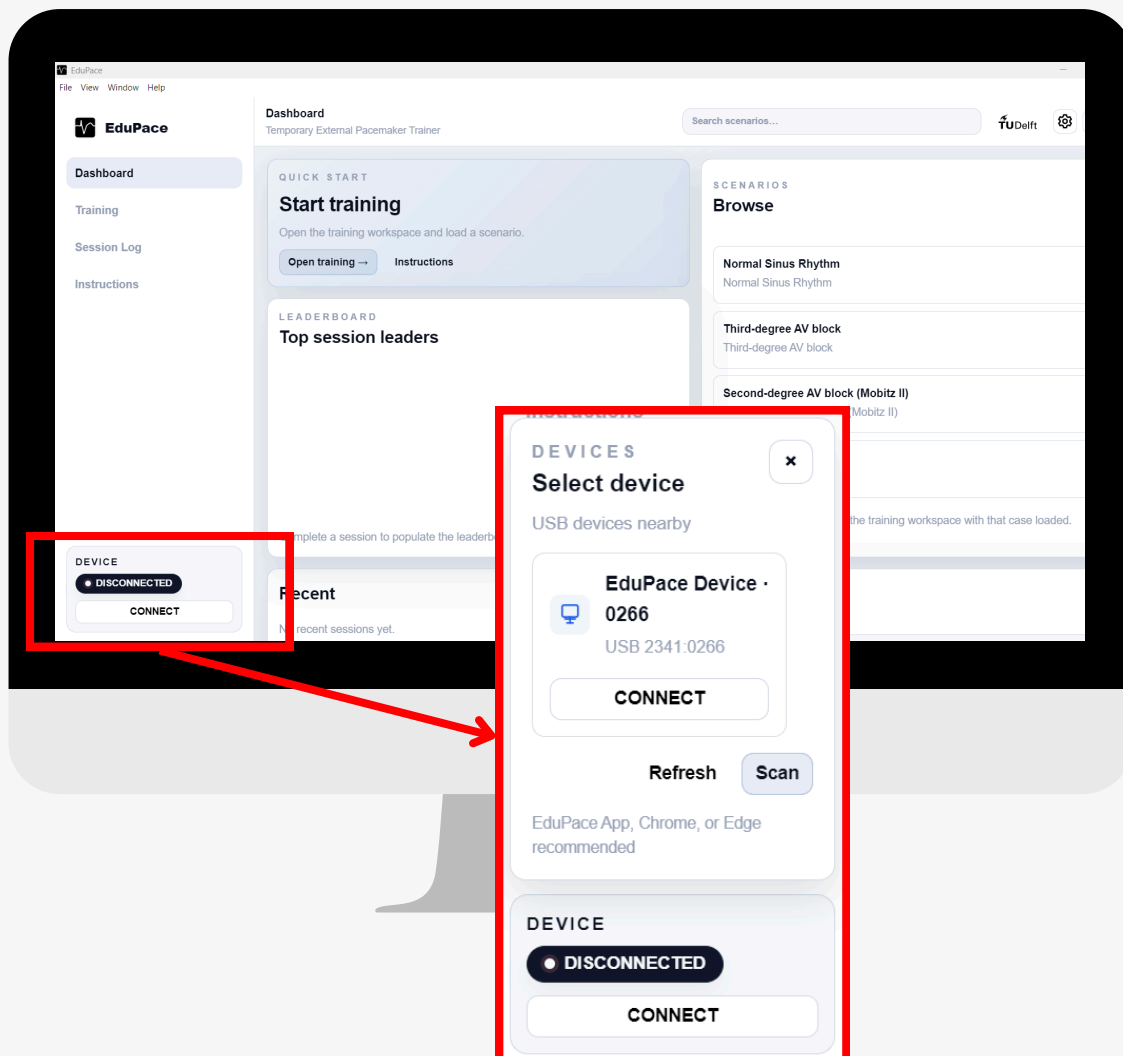
System Connector Setup

1. Connect a USB-C cable to the EduPace device and plug the other end into your computer.
2. Press START on the EduPace device and then press the power button to turn it on.



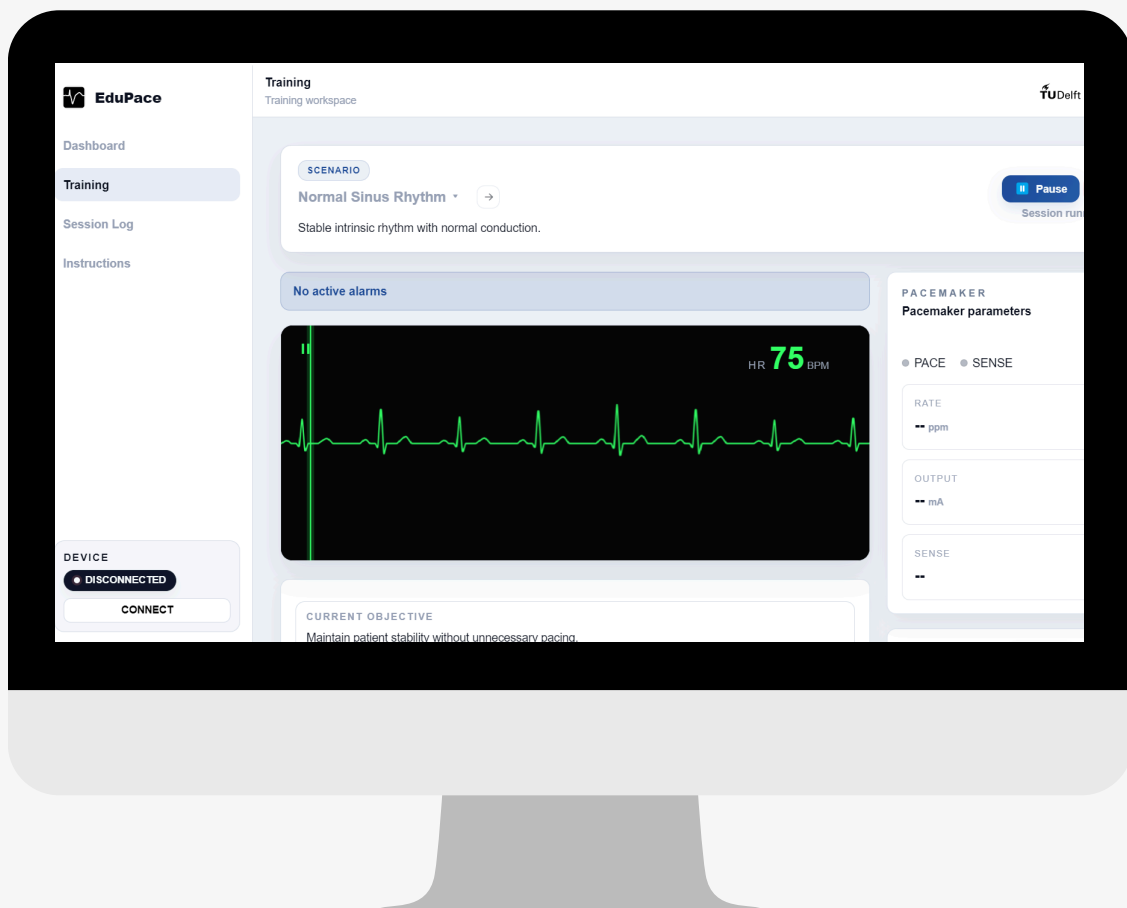
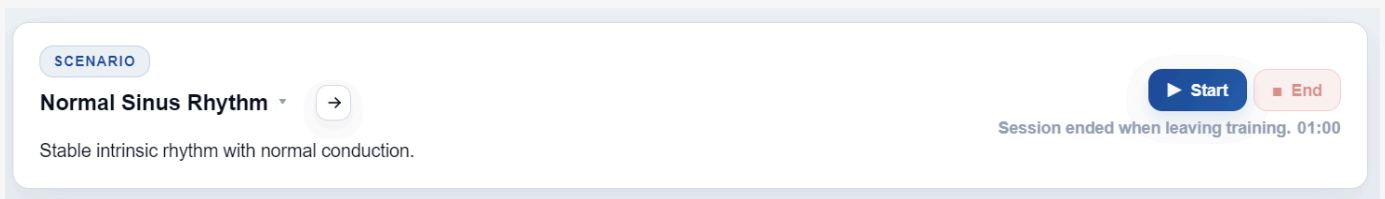
System Setup

3. Open the EduPace software and click Connect. When the list of available devices appears, choose the EduPace device and then click Connect to establish the connection.



System Setup

- Go to Training and select a scenario and then click "Start" to begin the session. The ECG trace will then appear on the screen.



To Turn On/Off

1. To start pacing, briefly press the On/Off button. When the device turns on, the power button lights up green and the background changes from gray to white.



2. To turn off the system, press and hold the On/Off button for more than 2 seconds.

Note: The system must be unlocked before powering off, similar to the Medtronic external pacemaker.

Lock/Unlock

1. To lock the device, briefly press the Lock/Unlock button. The button will turn yellow, indicating that the device is locked. While locked, rotating the RATE, OUTPUT, or SENSE dials will have no effect.



Note: If the RATE, OUTPUT, or SENSE values are not changed for 60 seconds, the system will automatically lock, similar to the Medtronic external pacemaker.

Rate, Output, and Sensitivity Adjustments

1. When EduPace is turned on, the start values are:

RATE: 80 ppm (range 30-200 ppm)

OUTPUT: 10 mA (range 0.1-25 mA)

SENSITIVITY: 2 mV (range 0.1-20 mV)



2. When the system is unlocked, rotate dials to adjust RATE, OUTPUT and SENSE using rotary dials. The left green LED flashes to indicate pacing activity, and the right blue LED flashes to indicate sensing activity.



EduPace

2025-2026